

Rescue Line

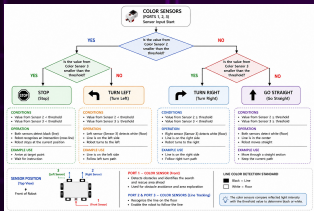


team:Bricks

2024 RoboCup Korea Rescue Line U19 2nd

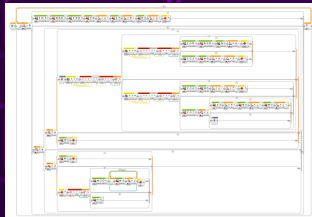
2026 RoboCup Korea Rescue Line U19 1st

flowchart



This program uses the ratio of red, blue, green, or RGB. It's a program that measures and passes through an intersection. If it's green. If the ratio is over 56 percent, the green sticker is bath. I make a U-turn after judging that it's attached. And if the ratio of green exceeds 56% on the right side, only on the right side. Decide that there's a green sticker on it and turn right. Let's do it. Finally, the green ratio is 56 percent on the left alone. If you flip it over, there's a green sticker on the left side. Make a judgment and make a left turn.

software



team member

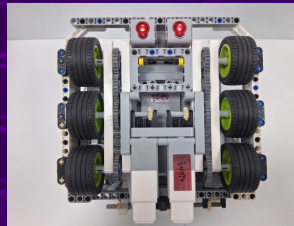


I made a wheel with 6 wheels to make it go up the hill well, I put 2 color sensors to get on the line, I put 1 color sensor for obstacle detection and evacuation zone detection. And I made an arm to catch the casualty.



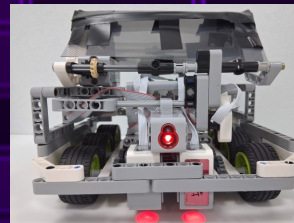
To get on the line well, I created a two-sensor line-tracing program, an RGB value measurement program for detecting green intersections, and an obstacle detection program.

hardware



This robot is not supposed to slip on the hill. The friction force was strengthened by using six wheels.

And instead of a large motor, a medium motor was used to reduce the size and weight of the robot and gear it to rotate accurately. And two color sensors were used on both sides to stabilize the line training.



sweeping one's arms for a ball. It's made in the form of a trash tray and for holding it up.

I made a basket for the body. And green. To distinguish between a refuge and a red refuge. I put a color sensor on it. And in the distress area. I made a guard to align the walls.